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Insights into the Solvation Characteristics of Zwitterionic Deep Eutectic Solvents Using Multiple Polarity Scales

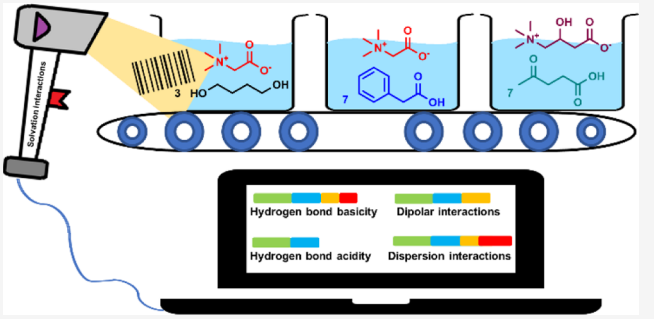
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ABSTRACT: Deep eutectic solvents (DESs) formed by mixing zwitterionic hydrogen bond acceptors (HBAs) with diol- and organic acid-based hydrogen bond donors (HBDs) are popular sustainable solvents. In addition to possessing an amphiphilic character, zwitterionic DESs offer unique solvation properties, unlike those composed of halide-containing HBAs. While all other popular classes of DESs have been investigated, no study has attempted to characterize the solvation characteristics of zwitterionic DESs. In this study, the solvation properties of betaine [Bet]-, sulfobetaine-, and L-carnitine [L-Car]-based DESs were thoroughly investigated for the first time using solvatochromic dye methods and inverse gas chromatography (IGC). The Nile red polarity scale revealed that DESs composed of organic acids were generally more polar than diol-based DESs. Kamlet−Taft solvent parameters revealed that glycerol-based DESs offered exceptionally high hydrogen bond donating ability compared to those composed of 1,4-butane diol and triethylene glycol. The Abraham solvation parameter model revealed hydrogen bond basicity to be the most dominant interaction, and it was generally higher for diol-based HBDs compared to organic acid-based HBDs. Additionally, sulfobetaine HBAs possessing longer alkyl chains generally offered higher dispersive-type interactions and hydrogen bond basicity compared to [Bet] and [L-Car] salts. This is the first study to combine and compare results from solvatochromic dye methods and IGC in an attempt to comprehensively explain and predict the solvation characteristics of zwitterionic DESs.



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1. INTRODUCTION

Deep eutectic solvents (DESs) are a popular choice of sustainable designer solvents employed in numerous scientific disciplines.[1,2] They are multicomponent fluids composed of a hydrogen bond acceptor (HBA) that interacts with a hydrogen bond donor (HBD) to yield eutectic mixtures possessing low melting points compared to the HBA and HBD alone.[3,4] While HBAs typically comprise tetraalkylammonium, phosphonium, imidazolium, and choline salts, the choice of HBDs can include organic acids, alcohols, amides, and carbohydrates.[5,6] DESs often possess desirable properties such as tunable viscosities,[1,7] low vapor pressures,[1,8] and ease of preparation,[1,9] and are preferentially employed in catalysis,[10,11] organic synthesis,[12] chemical separations,[13,14] and electrochemical applications.[15]

Several classes of DESs have been established over the years, each possessing unique features and characteristics.[1] Among these is the zwitterionic class of DESs that are predominantly composed of betaine [Bet] and L-carnitine [L-Car] salts employed as HBAs that offer lower cytotoxicity and do not generate undesirable side products compared to tetraalkylammonium and choline salts.[16,17] It has been reported that halogen counterions can impart strong nucleophilic, alkaline, and oxidizing properties to halide-containing DESs that can severely corrode steel equipment employed in industrial plants.[17−19] More importantly, betaine is an industrially obtained renewable resource formed as a byproduct of sugar production, while choline chloride [Ch^+][Cl^-] is acquired from fossil fuels.[20,21] Additionally, zwitterionic surfactants possess amphiphilic properties that allow them to form micelle-like assemblies and behave as hydrophilic and hydrophobic nanocontainers, properties and characteristics that are not achieved by other classes of DESs.[22,23]

Despite being capable of forming DESs with fewer types of HBDs compared to [Ch^+][Cl^-] and tetraalkylammonium salts, zwitterionic HBAs can produce eutectic mixtures that can target niche applications.[24,25] For example, the [Bet]/2 triethylene glycol DES mixed with 30% (v/v) of water was found to be more efficient in extracting chlorogenic acids from spent coffee grounds (approximately 4.75 mg g^{-1} of sample) compared to several [Ch^+][Cl^-] DESs when employed at a sample-to-solvent ratio of 1:15 (g/mL).[23] It has been argued that the amphiphilic nature of betaine salts can enhance their

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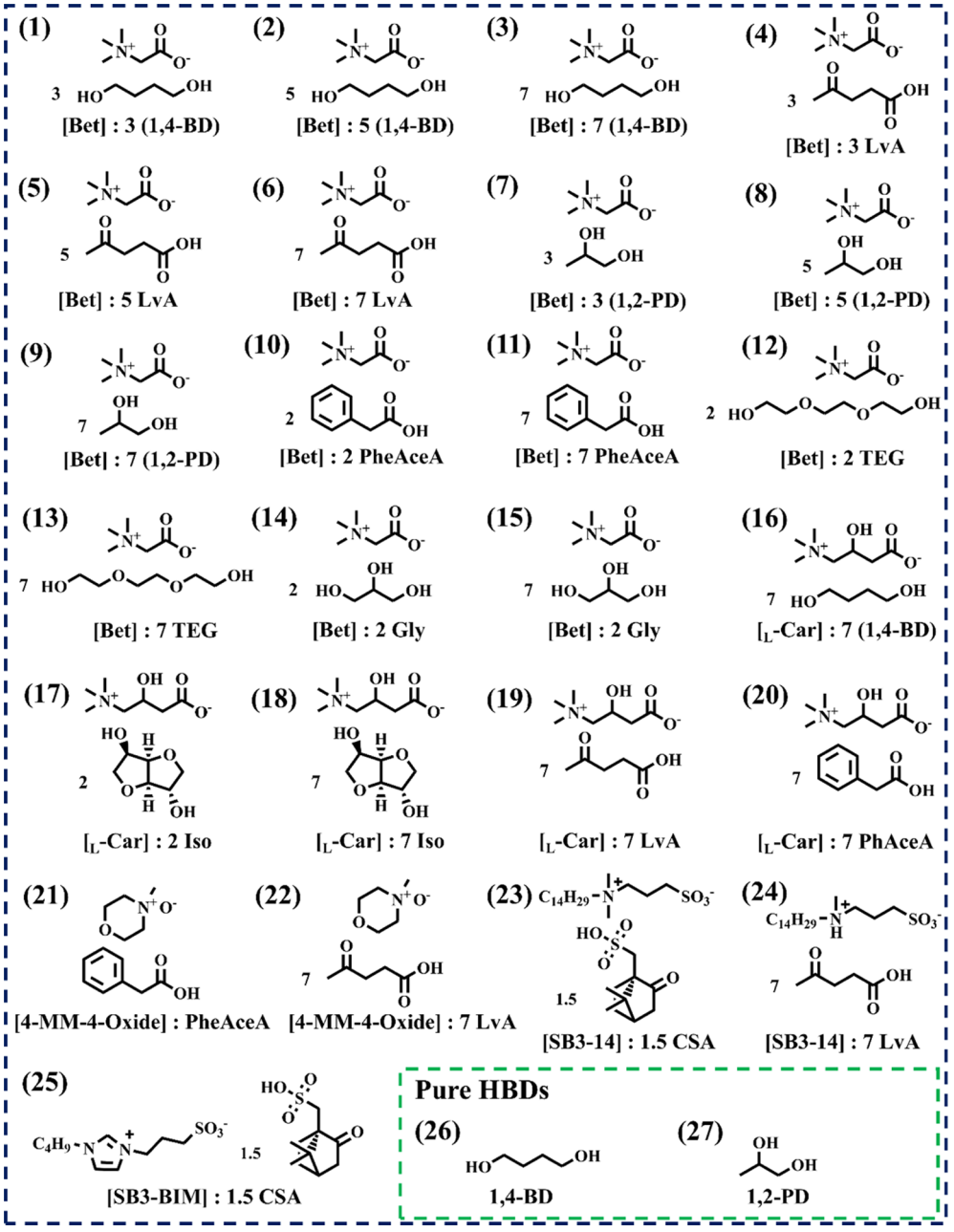


Figure 1. Chemical structures, numbering scheme, and abbreviations used for the DESs examined in this study. Abbreviations: triethylene glycol (TEG); 1,2-propane diol (1,2-PD); 1,4-butane diol (1,4-BD); levulinic acid (LvA); isosorbide (Iso); glycerol (Gly); phenylacetic acid (PheAceA); (1S)-(+)-10-Camphorsulfonic acid (CSA); betaine ([Bet]); and L-carnitine ([L-Car]).

interactions with certain types of analytes compared to $[Ch^+][Cl^-]$, despite similarities in their chemical structures.[21,23] Previously, Bryant et al. highlighted that the presence of a carboxylate substituent and the absence of hydroxyl functional groups, as well as their surfactant-like nature in betaine HBAs compared to $[Ch^+][Cl^-]$, is responsible for varied electrostatic interactions,[26] thereby imparting differences in their solvation properties. Like all other classes of DESs, the polarity of zwitterionic DESs is strongly influenced by the type of HBA and HBD and/or their molar ratio.[18,27,28] In an attempt to understand their molecular interactions, Abranches et al. studied the solid−liquid phase diagrams of betaine-based DESs and found that betaine possesses a charge imbalance and cannot establish strong ion−ion interactions with itself, unlike salts that possess counter-ions.[20] Nevertheless, this study did not quantify the individual solvation interactions of these DESs.

Conventionally, the solvation properties of DES systems have been investigated by studying their solvatochromic response to construct the betaine dye scale (E_TN).[29,30] For example, an E_TN value of 0.36 was obtained for the $[Ch^+][Cl^-]$/glycolic acid DES compared to 0.79 for the $[Ch^+][Cl^-]$/malonic acid DES, thereby revealing their different solvation characteristics.[31] Given that the E_TN scale is dependent on a single dye measurement, the Kamlet−Taft solvent parameters employ several solvatochromic probes and can measure the solvent's hydrogen bond donating ability (α), hydrogen bond accepting ability (β), and dipolarity/polarizability (π^*).[31,32] While the fluorescence-based probe approach was the first to characterize DES solvation proper-

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ties,[33,34] it has failed in some cases to deconvolute these interactions. For example, Kim and co-workers reported a statistically similar $E_T N$ value of 0.843 for the $[Ch^+][Cl^-]$/urea/glycerol ternary DES system compared to 0.845 for the $[Ch^+][Cl^-]$/glycerol DES,[35] despite significant differences in their chemical structures. Similarly, Pandey et al. obtained $E_T N$ values of 0.82 for both the $[Ch^+][Cl^-]$/2 ethylene glycol DES and ethylene glycol,[36] demonstrating very little difference between the neat HBD and its DES. It has been concluded that the solvatochromic approach is insensitive to DES compositional changes due to their high polarity and is generally more reliable when the solvation properties from two different classes of DESs are compared.[31,32]

In an attempt to enhance the sensitivity in characterizing DES solvation properties, inverse gas chromatography (IGC) has been recently employed to investigate solute−solvent interactions.[34,37] The chromatographic retention of a diverse variety of probe molecules is measured on DES stationary phases as a function of temperature in order to characterize and quantify their unique and simultaneous interactions with DESs. To date, popular DES combinations comprising tetraalkylammonium, phosphonium, and choline salts have been studied using both the IGC[38] and the solvatochromic dye approaches.[30,31] While zwitterionic-based DESs composed of different classes of HBDs have been employed in several applications, no study has attempted to thoroughly characterize their individual solvation interactions. Currently, only a few zwitterionic DESs comprising *N*-oxide HBAs have been screened using the Nile red solvatochromic scale,[22] which only provides a single value for polarity and does not explain all types of unique interactions that the DESs are capable of undergoing.[2] Consequently, the solvation properties of zwitterionic DESs have not been investigated with either Kamlet−Taft parameters or IGC and, thus, cannot be compared to those of other DES classes.

In this study, zwitterionic DESs formed by mixing betaine-, L-carnitine, and sulfobetaine-based HBAs with alcohol- and organic acid-based HBDs were characterized for the first time using both the IGC and the solvatochromic dye method. The choice of HBA and HBD was carefully made to incorporate unique functional groups, such as phenyl and imidazolium substituents, while the molar ratio of HBD was systematically modulated to investigate the effect of the HBA/HBD ratio. Additionally, the solvation properties of DESs were benchmarked against neat HBDs that comprise them. Two different solvatochromic scales (Nile red polarity scale and the Kamlet−Taft solvent parameters) were employed to thoroughly characterize the solvation properties of all DESs using solvatochromic dyes. This study demonstrates the need to simultaneously employ multiple types of approaches and empirical polarity scales in order to fully characterize the solvation properties of complex DESs and is the first to thoroughly investigate subtle differences in solvation characteristics between surfactant-like and halide-containing DESs.

2. EXPERIMENTAL SECTION

2.1. Reagents, Materials, DES Preparation and Characterization. A comprehensive list of reagents and starting materials employed in this study is provided in the Supporting Information. All HBAs and HBDs were thoroughly dried for 48 h in a vacuum oven maintained at 25 °C prior to being employed in DES preparation using the grinding method. Precise quantities of reagents corresponding to desired HBA/HBD ratios (see Figure 1) were weighed into a ceramic mortar using a clean spatula for zwitterionic salts and a transfer pipette for liquid HBDs. Contents of the mortar were ground together with a pestle until a homogeneous liquid was obtained. The newly prepared DESs were transferred into 20 mL glass vials and their purity was characterized using nuclear magnetic resonance (NMR) spectroscopy prior to further analysis; all ^{13}C and ^{1}H NMR spectra are provided in the Supporting Information. The water content of each DES was found to be under 0.9 wt % (Table S1), as determined using a Metrohm 831 Karl Fischer coulometric titrator by measuring each sample in triplicate. Additionally, DES viscosity measurements were performed at 21.6 °C using an AMETEK Brookfield DV1 cone and plate viscometer (Middleborough, MA, USA) equipped with a CPA-51Z cone spindle and compared to halide-containing DESs that had similar chemical structures; data are provided in Table S2.

Thermal properties of DESs were characterized using a NETZSCH DSC 214 Polyma instrument (Selb, Germany). DESs that possessed a liquid consistency at room temperature were cooled from 40 to −120 °C at a rate of 10 °C min^{-1} and subsequently heated to 80 °C at the same rate, while those that appeared as solids under ambient conditions were cooled to −65 °C at a rate of 10 °C min^{-1} before reheating them to 200 °C at the same rate. DSC thermograms were analyzed using NETZSCH Proteus software to obtain the melting point and/or glass transition temperatures for all DESs, as shown in Table S3.

2.2. Solvatochromic Dye Measurements. To conduct polarity measurements for zwitterionic DESs using the solvatochromic approach, Nile red (NR), 4-nitroaniline (4-NA), and *N*,*N*-diethyl-4-nitroaniline (*N*,*N*-DENA) dyes were employed.[29] The NR and *N*,*N*-DENA dyes were dissolved in chloroform to prepare 0.01 M stock solutions, while 4-NA was solubilized in methanol at the same concentration. Dye solutions were stored under 4 °C, and 100 μL of each dye was pipetted into clean Eppendorf tubes. The solvent was evaporated by passing a gentle stream of nitrogen, leaving behind pure dyes as residues in the tubes. A 100 μL volume of DESs was placed into the Eppendorf tubes and thoroughly mixed with each of these dyes to fully solubilize them in the eutectic mixtures. A small volume of each DES/dye solution (20 μL) was transferred into a 96-well plate and analyzed using a Synergy H1 microplate reader (Biotek, Winooski, Vermont). Spectral scanning was performed between wavelengths of 300 and 700 nm in 1 nm increments. The maximum absorption wavelength (λ_{max}) was recorded for each dye, and representative UV−Vis spectra can be found in Figure S1 of the Supporting Information.

2.3. DES Chromatographic Column Preparation and Characterization. DESs were employed as GC stationary phases using the static method of column preparation. A 32 mg mass of each DES was placed inside a 20 mL glass vial and dissolved in a mixture of dichloromethane/methanol (up to 5% methanol) to prepare DES coating solutions at a concentration of 0.32% w/v. Five-meter segments of deactivated fused-silica capillary were filled with DES coating solutions and sealed at one end using a metal clamp. The other end was connected to a vacuum and placed inside a water bath maintained at 40 °C. The solvent was steadily evaporated, and the DESs were coated onto the inner walls of the silica capillary. All DESs were coated as a uniform film with a thickness of 0.20 μm.[39]

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

The newly prepared DES columns were placed inside two Agilent 7890B gas chromatographs and conditioned at 60 °C for 30 min under a 1 mL/min constant flow of helium. Inlet/detector temperatures were maintained at 150 °C, while the GC oven temperature was modulated between 30 and 50 °C. Additionally, the inlet split ratio was maintained at 20:1. Columns that were prepared from DESs composed of 1,2-PD and PheAceA HBDs had both of their ends connected to 10 cm segments of empty fused silica capillary using press-fit column connectors for GC (Agilent Technologies, Santa Clara, USA), which were then connected to the inlet/detector. Column efficiencies were determined by separating naphthalene (1 mg/mL) at 60 °C and were found to vary between 1700 and 3600 plates/m. In addition, probe molecules comprising alcohols, haloalkanes, and nitroalkanes were solubilized in dichloromethane at a concentration of 1000 ppm (1 mg/mL), and 1 μL was injected onto the DES columns. A comprehensive list of 50 probe molecules employed in this study is provided in Table S4. Multiple linear regression analysis (MLRA) and statistical calculations were performed using Analyze-it (Microsoft, USA).

3. RESULTS AND DISCUSSION

Natural deep eutectic solvents (NADES), which are perceived to be the least toxic and most sustainable class of DESs, can be categorized into ionic and non-ionic. While non-ionic NADES are predominantly composed of long-chain fatty acids and menthol/thymol derivatives, the ionic class can be classified into zwitterionic and halogen-containing types. Given the unique structural variations of NADES, the two components of zwitterionic DESs were carefully chosen to represent combinations popularly employed in extractions, catalysis, and organic synthesis, as shown in Figure 1. Even though some eutectic mixtures may deviate significantly from the eutectic point,[40] HBA and HBD combinations examined in this work have been described as DESs in previous studies;[16,22,41] they are being referred to as DESs for the sake of simplicity. Most DESs comprised the [Bet] HBA, as it offered the closest resemblance to the $[Ch^+][Cl^-]$ salt by substituting the hydroxyl functional group with a carboxylate substituent, which also acts as the primary hydrogen bond base in the zwitterion. In contrast, the [L-Car] salt incorporates features from the [Bet] and $[Ch^+][Cl^-]$ HBAs by possessing both the hydroxyl and the carboxylate functionalities. The type of hydrogen base alone can significantly influence the solvation properties as $[Ch^+][Cl^-]$ is known to undergo stronger hydrogen bonding between the hydroxyl functional group of its cation and chloride anion. In contrast, the [Bet] HBA cannot form strong interactions between its shielded quaternary nitrogen and the carboxylate substituent.[20] Additionally, [4-MM-4-oxide] was investigated to elucidate the effect of cyclic zwitterions, while [SB3-14] and [SB3-BIM] were chosen to study the effect of carboxylate and sulfonate moieties. The diverse range of HBDs employed in this study included both straight-chained and branched organic acids and diols. Neat HBDs were also investigated to compare their solvation properties with those of DESs comprising them. In addition, most HBAs and HBDs were mixed together in 2−3 molar ratios to also study the effect of the HBD molar ratio.

3.1. Nile Red Dye Polarity Scale (E_{NR}). Nile red dye has been widely used to estimate the empirical polarity of solvents (their overall solvation ability),[29] as its λ_{max} has been shown to deviate by 100 nm between water (hydrophilic) and pentane (hydrophobic).[22] Given that it exhibits a positive solvatochromic shift, its λ_{max} is generally observed at higher wavelengths in polar solvents, leading to lower E_{NR} values when compared to non-polar ones.[22] Table 1 provides the E_{NR} values for 20 DESs composed of [Bet], [L-Car], [4-MM-4-oxide], and [SB3-14] salts as HBAs. The E_{NR} values ranged from (48.46 ± 0.10) to (51.24 ± 0.04), and they did not vary drastically, as changes in the dye's λ_{max} values are marginal in hydrogen bonding solvents.[29] The lowest polarity of (51.24 ± 0.04) was obtained for the [4-MM-4-oxide]/PheAceA DES, which is comparable to a reported value of 50.94.[22] In contrast, the [Bet]/2 Gly and [Bet]/7 Gly DESs offered among the highest polarities of (48.58 ± 0.04) and (48.46 ± 0.10), respectively, as the addition of multiple hydroxyl functional groups makes the

Table 1. E_{NR} Polarity Values and Kamlet−Taft Solvent Parameters for DESs and Neat HBDs Comprising [Bet], [L-Car], [4-MM-4-Oxide], and [SB3-14] Salts as HBAs and TEG, Gly, Iso, PheAceA, LvA, CSA, 1,4-BD, and 1,2-PD HBDs[a]

DES	E_{NR} (kcal/mol)	α	β	π^*
[Bet]/3 (1,4-BD)	50.10 (0.04)	0.88 (0.01)	0.41 (0.01)	1.01 (0.01)
[Bet]/5 (1,4-BD)	50.13 (0.04)	0.88 (0.02)	0.43 (0.01)	0.99 (0.02)
[Bet]/7 (1,4-BD)	50.27 (0.11)	0.86 (0.01)	0.42 (0.01)	1.03 (0.01)
[Bet]/3 LvA	50.39 (0.04)	0.81 (0.01)	0.30 (0.01)	1.03 (0.01)
[Bet]/5 LvA	50.07 (0.02)	0.87 (0.02)	0.29 (0.01)	1.02 (0.02)
[Bet]/7 LvA	49.86 (0.04)	0.92 (0.01)	0.27 (0.01)	1.01 (0.01)
[Bet]/3 (1,2-PD)	50.25 (0.07)	0.84 (0.02)	0.60 (0.02)	1.00 (0.02)
[Bet]/5 (1,2-PD)	50.51 (0.03)	0.80 (0.02)	0.61 (0.02)	1.01 (0.02)
[Bet]/7 (1,2-PD)	50.30 (0.10)	0.83 (0.02)	0.59 (0.01)	1.00 (0.02)
[Bet]/7 TEG	50.72 (0.11)	0.73 (0.02)	0.29 (0.01)	0.98 (0.01)
[Bet]/2 Gly	48.58 (0.04)	1.11 (0.01)	0.49 (0.01)	1.16 (0.01)
[Bet]/7 Gly	48.46 (0.10)	1.13 (0.02)	0.47 (0.01)	1.17 (0.01)
[L-Car]/7 (1,4-BD)	50.07 (0.05)	0.88 (0.02)	0.36 (0.01)	1.02 (0.01)
[L-Car]/2 Iso	49.94 (0.04)	0.82 (0.03)		1.15 (0.01)
[L-Car]/7 Iso	49.90 (0.07)	0.85 (0.02)		1.14 (0.02)
[L-Car]/7 LvA	49.90 (0.07)	0.94 (0.02)	0.39 (0.01)	0.98 (0.01)
[L-Car]/7 PheAceA	50.56 (0.05)	0.77 (0.01)		1.02 (0.01)
[4-MM-4-oxide]/PheAceA	51.24 (0.04)	0.73 (0.01)	0.79 (0.01)	0.85 (0.01)
[4-MM-4-oxide]/7 LvA	49.93 (0.04)	0.96 (0.01)	0.42 (0.02)	0.95 (0.01)
[SB3-14]/7 LvA	49.81 (0.07)	1.03 (0.02)	0.49 (0.02)	0.87 (0.01)

[a]See Figure 1 for names, abbreviations, and chemical structures of all HBAs and HBDs. The λ_{max} for solvatochromic dyes used to calculate the E_{NR} values and Kamlet−Taft solvent parameters were measured in triplicate, and the standard deviation for each term is provided in parenthesis.

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

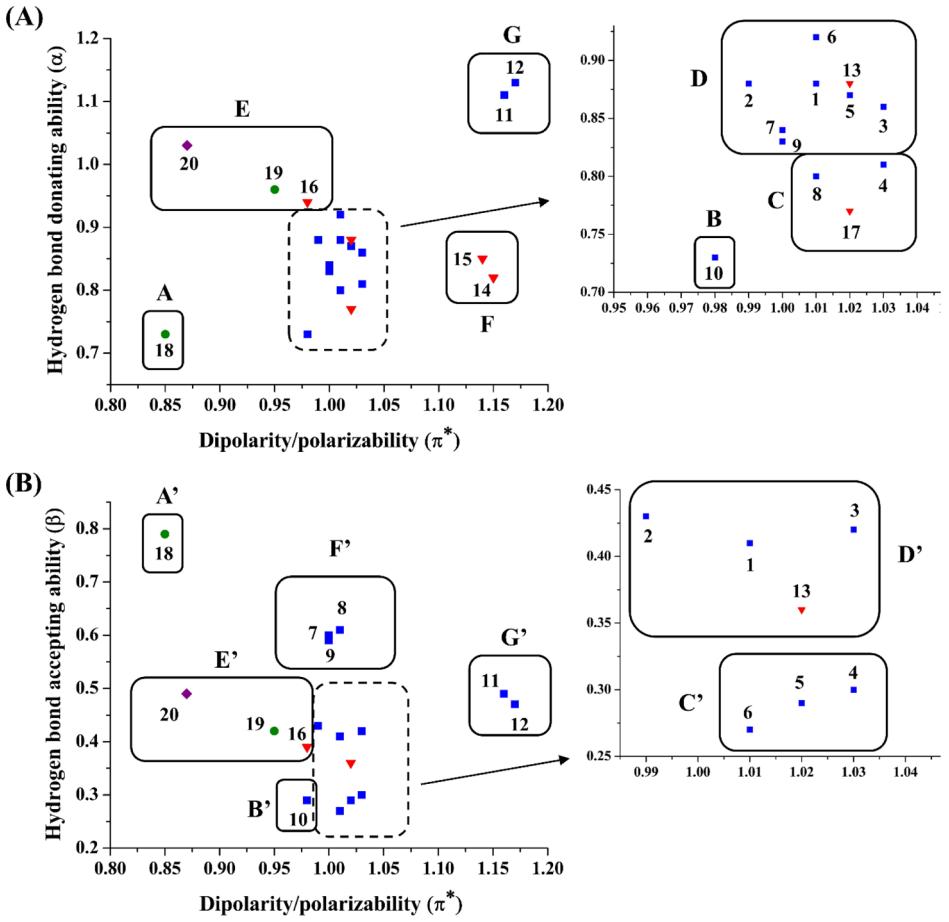


Figure 2. Plot illustrating the relation of (A) hydrogen bond donating ability versus dipolar interactions and (B) hydrogen bond accepting ability versus dipolar interactions at 25 °C for the 20 DESs evaluated in this study and grouped together using k-means clustering. The λ_{max} of *N*,*N*-DENA was used to calculate the π^*-term, NR and *N*,*N*-DENA to measure the α-term, and 4-NA and *N*,*N*-DENA to investigate the β-term. [Bet] DESs are represented by (blue ■), [L-Car] DESs are denoted by (red ▼), [4-MM-4-oxide] DESs are shown by (green ●), and [SB3-14] DESs are marked by (purple ◆). **1**, [Bet]/3 (1,4-BD); **2**, [Bet]/5 (1,4-BD); **3**, [Bet]/7 (1,4-BD); **4**, [Bet]/3 LvA; **5**, [Bet]/5 LvA; **6**, [Bet]/7 LvA; **7**, [Bet]/3 (1,2-PD); **8**, [Bet]/5 (1,2-PD); **9**, [Bet]/7 (1,2-PD); **10**, [Bet]/7 TEG; **11**, [Bet]/2 Gly; **12**, [Bet]/7 Gly; **13**, [L-Car]/7 (1,4-BD); **14**, [L-Car]/2 Iso; **15**, [L-Car]/7 Iso; **16**, [L-Car]/7 LvA; **17**, [L-Car]/7 PheAceA; **18**, [4-MM-4-oxide]/PheAceA; **19**, [4-MM-4-oxide]/7 LvA; and **20**, [SB3-14]/7 LvA.

DESs more polar.[41] Compared to the [Bet]/7 (1,4-BD) (50.27 ± 0.11), [Bet]/7 (1,2-PD) (50.30 ± 0.05), and [Bet]/7 TEG (50.72 ± 0.11) DESs, the [Bet]/7 LvA DES offered a much lower E_{NR} value of (49.86 ± 0.04), demonstrating the higher polarity of DESs composed of organic acids versus diols. In most cases, the polarity did not vary significantly when the molar ratio of DESs was varied, except for DESs composed of the levulinic acid-based HBD. The type of HBD appeared to have the greatest influence on polarity as similar values of (49.90 ± 0.07), (49.93 ± 0.04), (49.81 ± 0.07), and (49.86 ± 0.04) were obtained for the [L-Car]/7 LvA, [4-MM-4-oxide]/7 LvA, [SB3-14]/7 LvA, and [Bet]/7 LvA DESs, respectively. In addition, an E_{NR} value of (48.58 ± 0.04) obtained for the [Bet]/2 Gly DES was significantly lower than that reported for the $[Ch^+][Cl^-]$/2 Gly DES (49.6),[41] demonstrating the higher polarity of betaine-based DESs.

3.2. Kamlet−Taft Solvent Parameters. To characterize and quantify unique interactions using the solvatochromic dye approach, all DESs were examined using the Kamlet−Taft solvent parameters. The strength of the HBA is represented by the hydrogen bond basicity, while the hydrogen bond acidity provides a measure of the hydrogen bond-donating ability of the HBD. Table 1 lists the values for the hydrogen bond donating ability (α), hydrogen bond accepting ability (β), and dipolarity/polarizability (π^*). The hydrogen bond donating ability varied between (0.73 ± 0.01) and (1.13 ± 0.02) and was found to be strongly influenced by the type of HBD. The [Bet]/2 Gly and [Bet]/7 Gly DESs offered among the highest α-term values of (1.11 ± 0.01) and (1.13 ± 0.02), respectively, while both the [Bet]/7 TEG and [4-MM-4-oxide]/PheAceA DESs possessed the lowest hydrogen bond donating ability of (0.73 ± 0.02). A comparison of the [Bet]/7 (1,4-BD) (0.86 ± 0.01), [Bet]/7 (1,2-PD) (0.83 ± 0.02), and [Bet]/7 LvA (0.92 ± 0.01) DESs revealed that those composed of organic acid-based HBDs generally offered higher hydrogen bond donating ability, while 1,4-BD resulted in slightly higher α-term values in relation to 1,2-PD. Unlike diols, the α-term was observed to vary significantly for DESs composed of organic acids when the molar ratio of HBD was varied, as respective values of (0.81 ± 0.01), (0.87 ± 0.02), and (0.92 ± 0.01) were obtained for the [Bet]/3 LvA, [Bet]/5 LvA, [Bet]/7 LvA DESs. Additionally, the [SB3-14] HBA was found to possess the highest hydrogen bond donating ability with a value of (1.03 ± 0.02) for the [SB3-14]/7 LvA DES compared to the [Bet]/7 LvA (0.92 ± 0.01), [L-Car]/7 LvA (0.94 ± 0.02), and [4-MM-4-oxide]/7 LvA (0.96 ± 0.01) DESs.

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

Table 2. System Constants Obtained from the Abraham Solvation Parameter Model for DESs Composed of [Bet], [L-Car], [4-MM-4-Oxide], [SB3-BIM], and [SB3-14] Salts as HBAs and Various HBDs (TEG, Gly, Iso, PheAceA, LvA, CSA, 1,4-BD, and 1,2-PD)[d]

			system constants								
no.	DES	temp.(°C)	c	e	s	a	b	l	n[a]	R^{2b}	F[c]
		30	−3.44 (0.07)	0 (0)	2.28 (0.09)	5.29 (0.14)	0.50 (0.12)	0.62 (0.01)	37	0.99	864
1	[Bet]/3 (1,4-BD)	40	−3.34 (0.06)	0 (0)	2.14 (0.07)	4.96 (0.11)	0.48 (0.09)	0.57 (0.01)	37	0.99	1292
		50	−3.39 (0.05)	0 (0)	2.13 (0.06)	4.76 (0.10)	0.44 (0.08)	0.54 (0.01)	37	0.99	1488
		30	−3.42 (0.06)	0 (0)	2.14 (0.07)	5.21 (0.11)	0.56 (0.09)	0.64 (0.01)	37	0.99	1399
2	[Bet]/5 (1,4-BD)	40	−3.45 (0.05)	0 (0)	2.09 (0.06)	4.91 (0.10)	0.58 (0.09)	0.60 (0.01)	37	0.99	1590
		50	−3.47 (0.05)	0 (0)	2.08 (0.07)	4.73 (0.11)	0.57 (0.09)	0.56 (0.01)	37	0.99	1172
	[Bet]/7 (1,4-BD)	30	−3.51 (0.07)	−0.09 (0.06)	2.15 (0.07)	5.30 (0.12)	0.57 (0.09)	0.70 (0.02)	37	0.99	1015
3		40	−3.80 (0.07)	−0.10 (0.06)	2.22 (0.07)	5.25 (0.13)	0.63 (0.10)	0.69 (0.01)	37	0.99	819
		50	−3.72 (0.08)	−0.11 (0.07)	2.14 (0.08)	4.99 (0.14)	0.56 (0.10)	0.64 (0.01)	37	0.99	893
	[Bet]/5 LvA	30	−3.74 (0.09)	0.22 (0.09)	2.45 (0.07)	3.54 (0.11)	0.57 (0.10)	0.51 (0.02)	32	0.99	392
5		40	−3.71 (0.08)	0.24 (0.09)	2.39 (0.07)	3.37 (0.11)	0.48 (0.09)	0.48 (0.01)	32	0.99	384
		50	−3.85 (0.08)	0.17 (0.09)	2.21 (0.09)	3.25 (0.11)	0.44 (0.10)	0.44 (0.01)	32	0.99	423
	[Bet]/7 LvA	30	−3.67 (0.10)	0.30 (0.09)	2.41 (0.12)	3.58 (0.11)	0.59 (0.11)	0.53 (0.01)	32	0.99	387
6		40	−3.76 (0.09)	0.21 (0.09)	2.38 (0.12)	3.49 (0.11)	0.49 (0.10)	0.49 (0.01)	32	0.99	472
		50	−3.75 (0.09)	0.16 (0.09)	2.09 (0.11)	3.26 (0.10)	0.56 (0.11)	0.45 (0.01)	32	0.99	363
	[Bet]/3 (1,2-PD)	30	−3.62 (0.06)	0 (0)	2.22 (0.07)	5.25 (0.12)	0.34 (0.10)	0.64 (0.01)	38	0.99	1175
7		40	−3.79 (0.06)	0 (0)	2.28 (0.07)	5.19 (0.11)	0.33 (0.09)	0.62 (0.01)	38	0.99	1336
		50	−3.60 (0.06)	0 (0)	2.18 (0.08)	4.88 (0.12)	0.30 (0.10)	0.55 (0.01)	38	0.99	998
		30	−3.64 (0.06)	0 (0)	2.14 (0.07)	5.18 (0.11)	0.54 (0.09)	0.66 (0.01)	38	0.99	1595
8	[Bet]/5 (1,2-PD)	40	−3.70 (0.06)	0 (0)	2.09 (0.07)	4.96 (0.11)	0.58 (0.09)	0.63 (0.01)	38	0.99	1259
		50	−3.76 (0.06)	0 (0)	2.08 (0.07)	4.77 (0.12)	0.57 (0.09)	0.60 (0.01)	38	0.99	1241
		30	−3.67 (0.06)	0 (0)	2.11 (0.07)	5.14 (0.11)	0.65 (0.09)	0.69 (0.01)	38	0.99	1474
9	[Bet]/7 (1,2-PD)	40	−3.80 (0.06)	0 (0)	2.10 (0.08)	4.99 (0.12)	0.64 (0.10)	0.66 (0.01)	38	0.99	1048
		50	−3.91 (0.06)	0 (0)	2.13 (0.07)	4.86 (0.12)	0.59 (0.09)	0.65 (0.01)	38	0.99	1168
		30	−3.34 (0.07)	−0.16 (0.08)	2.31 (0.07)	3.82 (0.11)	0.56 (0.09)	0.67 (0.01)	37	0.99	694
10	[Bet]/2 PheAceA	40	−3.38 (0.06)	−0.12 (0.07)	2.24 (0.06)	3.69 (0.11)	0.41 (0.08)	0.63 (0.01)	37	0.99	739
		50	−3.21 (0.05)	−0.09 (0.07)	2.16 (0.06)	3.53 (0.10)	0.32 (0.08)	0.58 (0.01)	37	0.99	717
		30	−3.25 (0.06)	−0.19 (0.07)	2.17 (0.08)	3.58 (0.10)	0.79 (0.09)	0.73 (0.02)	37	0.99	797
11	[Bet]/7 PheAceA	40	−3.27 (0.06)	−0.14 (0.06)	2.06 (0.07)	3.33 (0.09)	0.72 (0.09)	0.68 (0.01)	37	0.99	854
		50	−3.23 (0.06)	−0.10 (0.06)	1.90 (0.08)	3.10 (0.12)	0.63 (0.10)	0.63 (0.01)	37	0.99	776
		30	−3.32 (0.06)	0.27 (0.06)	2.39 (0.07)	4.49 (0.11)	0 (0)	0.60 (0.02)	37	0.99	892
13	[Bet]/7 TEG	40	−3.34 (0.05)	0.22 (0.06)	2.24 (0.07)	4.31 (0.11)	0 (0)	0.57 (0.02)	37	0.99	921
		50	−3.10 (0.06)	0.15 (0.06)	2.13 (0.07)	4.09 (0.12)	0 (0)	0.52 (0.01)	36	0.99	924
		30	−3.37 (0.09)	0 (0)	2.47 (0.10)	4.59 (0.16)	0.65 (0.16)	0.34 (0.02)	35	0.99	501
15	[Bet]/7 Gly	40	−3.24 (0.08)	0 (0)	2.34 (0.10)	4.42 (0.16)	0.62 (0.15)	0.28 (0.02)	35	0.99	452
		50	−3.07 (0.08)	0 (0)	2.23 (0.09)	4.26 (0.15)	0.61 (0.14)	0.24 (0.02)	35	0.99	406
		30	−3.64 (0.06)	0 (0)	2.19 (0.07)	5.21 (0.11)	0.78 (0.09)	0.66 (0.01)	37	0.99	1614
16	[L-Car]/7 (1,4-BD)	40	−3.70 (0.05)	0 (0)	2.15 (0.06)	5.18 (0.11)	0.59 (0.08)	0.64 (0.01)	37	0.99	1631
		50	−3.58 (0.05)	0 (0)	2.07 (0.06)	4.78 (0.10)	0.63 (0.08)	0.58 (0.01)	37	0.99	1696
		30	−3.48 (0.07)	0 (0)	2.68 (0.10)	4.80 (0.15)	0.46 (0.13)	0.52 (0.02)	35	0.99	661
18	[L-Car]/7 Iso	40	−3.49 (0.07)	0 (0)	2.56 (0.09)	4.61 (0.14)	0.47 (0.12)	0.48 (0.02)	35	0.99	710
		50	−3.47 (0.06)	0 (0)	2.49 (0.08)	4.14 (0.14)	0.59 (0.11)	0.47 (0.01)	35	0.99	804
		30	−3.56 (0.14)	0.28 (0.15)	2.23 (0.16)	3.98 (0.25)	0.69 (0.23)	0.52 (0.04)	32	0.98	385
19	[L-Car]/7 LvA	40	−3.52 (0.11)	0.21 (0.12)	2.16 (0.13)	3.58 (0.20)	0.65 (0.18)	0.48 (0.03)	32	0.98	417
		50	−3.75 (0.09)	0.16 (0.09)	2.09 (0.11)	3.26 (0.10)	0.56 (0.11)	0.45 (0.01)	32	0.99	463
		30	−3.81 (0.07)	−0.26 (0.08)	2.17 (0.09)	3.77 (0.14)	0.64 (0.12)	0.69 (0.02)	36	0.99	915
20	[L-Car]/7 PheAceA	40	−3.56 (0.06)	−0.19 (0.07)	2.11 (0.08)	3.54 (0.12)	0.51 (0.10)	0.64 (0.02)	36	0.99	839
		50	−3.68 (0.07)	−0.11 (0.07)	2.03 (0.09)	3.35 (0.13)	0.59 (0.12)	0.59 (0.02)	36	0.99	823
	[4-MM-4-oxide]/PheAceA	30	−3.32 (0.05)	−0.16 (0.05)	2.61 (0.07)	4.61 (0.11)	0 (0)	0.68 (0.01)	36	0.99	1383
21		40	−3.32 (0.05)	−0.14 (0.05)	2.53 (0.07)	4.39 (0.11)	0 (0)	0.64 (0.01)	36	0.99	1308
		50	−3.27 (0.05)	−0.08 (0.05)	2.39 (0.07)	4.13 (0.09)	0 (0)	0.59 (0.01)	36	0.99	1243
	[4-MM-4-oxide]/7 LvA	30	−3.08 (0.07)	0 (0)	2.42 (0.09)	4.01 (0.13)	0.79 (0.10)	0.70 (0.01)	35	0.99	882
22		40	−3.09 (0.06)	0 (0)	2.35 (0.07)	3.85 (0.11)	0.77 (0.09)	0.65 (0.01)	35	0.99	1119
		50	−3.37 (0.05)	0 (0)	2.27 (0.06)	3.61 (0.10)	0.68 (0.08)	0.59 (0.01)	36	0.99	1292
		30	−3.39 (0.06)	0 (0)	2.35 (0.07)	4.52 (0.12)	0 (0)	0.67 (0.01)	36	0.99	923
23	[SB3-14]/1.5 CSA	40	−3.50 (0.06)	0 (0)	2.32 (0.07)	4.46 (0.12)	0 (0)	0.65 (0.01)	36	0.99	965
		50	−3.43 (0.06)	0 (0)	2.20 (0.07)	4.15 (0.11)	0 (0)	0.60 (0.01)	36	0.99	1024

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

Table 2. continued

no.	DES	temp.(°C)	system constants								
			c	e	s	a	b	l	n^a	R^{2b}	F^c
		30	−3.08 (0.07)	−0.23 (0.06)	2.20 (0.09)	4.03 (0.13)	0.57 (0.10)	0.70 (0.01)	36	0.99	1196
24	[SB3-14]/7 LvA	40	−3.09 (0.06)	−0.17 (0.06)	2.08 (0.07)	3.84 (0.11)	0.45 (0.09)	0.67 (0.01)	36	0.99	1229
		50	−3.20 (0.05)	−0.09 (0.05)	1.99 (0.06)	3.62 (0.10)	0.34 (0.08)	0.66 (0.01)	36	0.99	1275
		30	−3.15 (0.06)	−0.24 (0.05)	1.92 (0.07)	4.40 (0.10)	−0.27 (0.09)	0.83 (0.01)	36	0.99	1531
25	[SB3-BIM]/1.5 CSA	40	−3.14 (0.05)	−0.19 (0.05)	1.83 (0.06)	4.06 (0.09)	−0.25 (0.08)	0.78 (0.01)	36	0.99	1680
		50	−3.15 (0.05)	−0.17 (0.05)	1.76 (0.06)	3.81 (0.09)	−0.26 (0.08)	0.74 (0.01)	36	0.99	1495
		30	−3.73 (0.08)	0 (0)	1.79 (0.10)	4.55 (0.14)	1.36 (0.12)	0.70 (0.02)	36	0.99	714
26	1,4-BD	40	−3.57 (0.06)	0 (0)	1.72 (0.08)	4.46 (0.14)	1.28 (0.12)	0.68 (0.01)	36	0.99	1099
		50	−3.63 (0.06)	0 (0)	1.70 (0.07)	4.28 (0.13)	1.25 (0.11)	0.66 (0.01)	36	0.99	1217
		30	−3.78 (0.09)	−0.24 (0.09)	1.93 (0.09)	4.92 (0.13)	0.89 (0.11)	0.76 (0.01)	36	0.99	681
27	1,2-PD	40	−3.62 (0.09)	−0.16 (0.08)	1.81 (0.09)	4.74 (0.16)	0.86 (0.12)	0.69 (0.01)	36	0.99	569
		50	−3.79 (0.07)	−0.13 (0.07)	1.90 (0.09)	4.56 (0.14)	0.91 (0.12)	0.67 (0.01)	36	0.99	666

[a] n, number of probes subjected to MLRA. [b] R^2, correlation coefficient. [c] F, Fisher F-statistic. [d] The names, abbreviations, and chemical structures of all HBAs and HBDs are provided in Figure 1.

The hydrogen bond accepting ability (β) was observed to vary significantly depending on the DES chemical structure, as values ranged from (0.27 ± 0.01) for the [Bet]/7 LvA DES to (0.79 ± 0.01) for the [4-MM-4-oxide]/PheAceA DES. DESs that generally offered higher hydrogen bond donating ability exhibited the lowest β-term values. Due to the strong dependence on pKa, organic acid-based DESs possessed much lower hydrogen bond accepting ability compared to those composed of diols, as observed from β-term values of (0.43 ± 0.01), (0.29 ± 0.01), and (0.61 ± 0.01) for the [Bet]/5 (1,4-BD), [Bet]/5 LvA, and [Bet]/5 (1,2-PD) DESs, respectively. A comparison of diol-based DESs revealed that those composed of 1,2-PD offered much higher hydrogen bond accepting ability than those formed with 1,4-BD. Additionally, values for the β-term did not vary significantly when the molar ratio of HBD was varied, regardless of the HBD. The nature of HBA appeared to drastically influence the hydrogen bond accepting ability, as observed by comparing the β-term values of the [SB3-14]:7 LvA (0.49 ± 0.02), [Bet]/7 LvA (0.27 ± 0.01), [L-Car]/7 LvA (0.39 ± 0.01), and [4-MM-4-oxide]/7 LvA (0.42 ± 0.02) DESs. Compared to a previously reported β-term value of 0.554 for the [Ch^+][Cl^-]/2 Gly DES,[31] the [Bet]/2 Gly DES exhibited a slightly lower hydrogen bond accepting ability of (0.49 ± 0.01), demonstrating a key difference between the surfactant-like and halide-containing classes of DES. In addition, the β-term for DESs composed of the isosorbide HBD could not be evaluated as no peak was detected in the UV−Vis wavelength region for the 4-NA dye.

The π^*-term represents all of the non-specific interactions that the DESs are capable of undergoing, including dipole−dipole and dipole-induced dipole interactions.[29] Dipolar interactions were found to range from (0.87 ± 0.01) to (1.17 ± 0.01) and did not vary drastically. The π^*-term values for betaine-based DESs composed of LvA, 1,4-BD, and 1,2-PD were found to be statistically similar at all molar ratios of HBD, indicating that the dipolarity/polarizability did not vary significantly between diols and organic acids. However, both of the [Bet]/2 Gly and [Bet]/7 Gly DESs exhibited significantly higher dipolar interactions of (1.16 ± 0.01) and (1.17 ± 0.01), respectively, which can be attributed to a larger number of hydrogen bond donating sites for glycerol compared to diols and LvA.[36] While the π^*-term values for the [Bet]/7 LvA (1.02 ± 0.01) and [L-Car]/7 LvA (0.98 ± 0.02) DESs were similar, the [SB3-14]:7 LvA DES exhibited a significantly lower value of (0.87 ± 0.01), demonstrating the lower dipolarity/polarizability of sulfobetaine-based DESs.

The studied DESs were further classified into groups based on the results from the Kamlet−Taft solvent parameters. Figure 2A shows values of hydrogen bond donating ability (α) plotted against dipolarity/polarizability (π^*) for 20 DESs. Using k-means clustering (see Supporting Information for details), it can be observed that the [4-MM-4-oxide]/PheAceA and [Bet]/7 TEG DESs exclusively formed clusters A and B, respectively. Cluster C is composed of both [Bet] (blue squares) and [L-Car] (red triangles) HBAs mixed with diols as well as organic acids, while cluster D is predominantly composed of diol-based HBDs. In contrast, cluster E is solely composed of DESs formed by the LvA HBD and [L-Car], [4-MM-4-oxide], and [SB3-14] HBA. DESs composed of Iso and Gly HBDs exclusively formed clusters F and G, respectively. While there appears to be no specific trend between the hydrogen bond donating ability and dipolar interactions, it can be observed that DESs possessing similar HBDs generally tend to fall in the same cluster, indicating that the solvation properties of betaine and sulfobetaine DES are more strongly influenced by the type of HBD than the HBA. A correlation of hydrogen bond accepting ability (β) with dipolarity/polarizability (π^*) is shown in Figure 2B. The same DESs fell into seven groupings using k-means clustering, and the hydrogen bond accepting ability also could not provide specific trends to dipolar interactions.

3.3. Comparing the [Bet], [L-Car], [SB3-BIM], [SB3-14], and [4-MM-4-Oxide] HBAs Using System Constants from Abraham Solvation Parameter Model. The rationale behind the choice of probe molecules and a description of all methods and equations used to obtain data for the solvatochromic dye methods and IGC are presented in the Supporting Information. Previously, GC inlet/detector temperatures for DES columns were optimized at 150 °C, while oven temperatures were kept below 70 °C in order to maintain the integrity of the DES stationary phase. However, chromatographic columns prepared with [Bet] and [L-Car]-based DESs composed of 1,2-PD and PheAceA HBDs were found to be unstable at inlet/detector temperatures of 150 °C. To overcome this, press-fit column connectors for GC were

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

employed to connect both ends of the columns with empty segments of capillary that were placed inside the inlet/detector to prevent the DES from being exposed to high inlet/detector temperatures.

DESs composed of straight-chained and branched zwitterionic salts as the HBAs and diols, carbohydrates, and organic acids as the HBDs (Figure 1) were characterized using IGC at 30, 40, and 50 °C. All models were statistically sound based on the magnitude of Fisher *F*-statistics and the high correlation coefficients (0.99) of the multiple linear regression fit, as shown in Table 2. All DESs exhibited significant dipolarity (*s*-term), hydrogen bond basicity (*a*-term), and dispersive-type (*l*-term) interactions. Dipolar interactions did not vary too drastically for most DESs and were found to range between (2.07 ± 0.06) and (2.49 ± 0.08) at 50 °C for the [Bet] and [L-Car]-based DESs, while the *s*-term was observed to be as low as (1.76 ± 0.06) for the [SB3-BIM]/1.5 CSA DES. In contrast, the sulfobetaine DESs were observed to possess significantly higher dispersive-type interactions compared to the [Bet] and [L-Car]-based DESs, as exemplified by the highest *l*-term value of (0.74 ± 0.01) for the [SB3-BIM]/1.5 CSA DES, which can be attributed to its longer alkyl substituents. Comparing the [SB3-BIM]/1.5 CSA (1.76 ± 0.06) and [SB3-14]/1.5 CSA (2.20 ± 0.07) DESs revealed that the dipolar interactions of sulfobetaine DESs demonstrate a greater variation and are observed to be higher for HBAs composed of an aliphatic cation versus an aromatic ring. However, an opposite trend was observed for dispersive-type interactions, where the *l*-term values of (0.74 ± 0.01) and (0.60 ± 0.01) were obtained for the [SB3-BIM]/1.5 CSA and [SB3-14]/1.5 CSA (2.20 ± 0.07) DESs, respectively.

For all DESs, the hydrogen bond basicity was found to be the most dominant type of solvation interaction, irrespective of the HBA. Values for the *a*-term at 50 °C exhibited a wide range from (3.25 ± 0.11) to (4.99 ± 0.14) while DESs comprising the [4-MM-4-oxide] HBA generally possessed higher hydrogen bond basicity. The [L-Car]/7 (1,4-BD) DES offered a slightly lower hydrogen bond basicity of (4.78 ± 0.10) compared to (4.99 ± 0.14) for the [Bet]/7 (1,4-BD) DES, which can be attributed to the presence of a hydroxyl function group in the [L-Car] HBA that can function as an additional HBD. It was also observed that the [SB3-14]/1.5 CSA DES exhibited a higher hydrogen bond basicity of (4.15 ± 0.11) compared to (3.81 ± 0.09) for the [SB3-BIM]/1.5 CSA, keeping in line with the previously observed trend where greater *a*-term values were obtained for DESs composed of bulky tetraalkylammonium cations containing longer alkyl substituents.[34] Most HBAs were able to form DESs with LvA HBD at a molar ratio of 1:7, thereby facilitating a convenient comparison of the effect of the type of HBA on DES solvation interactions. A graphic representation of variations in the hydrogen bond basicity, dispersive-type interactions, and dipolarity/polarizability for 23 DESs can be found in Figure S2 for a visual comparison.

3.4. Effect of HBD on DES Solvation Interactions Measured Using the Abraham Solvation Parameter. DESs composed of the [Bet], [L-Car], [SB3-BIM], [SB3-14], and [4-MM-4-oxide] HBAs and diols, carbohydrates, and organic acids as HBDs were studied to examine the effect of HBD on solvation interactions. Lower hydrogen bond basicity was generally observed for DESs composed of organic acid HBDs, as exemplified by the lowest *a*-term value of (3.25 ± 0.11) obtained at 50 °C for the [Bet]/5 LvA DES. Similarly, dispersive-type interactions for the [Bet]/7 LvA DES (0.45 ± 0.01) were also significantly low compared to (0.64 ± 0.01) and (0.65 ± 0.01) for the [Bet]/7 (1,4-BD) and [Bet]/7 (1,2-PD) DESs, respectively.

The [Bet] and [L-Car] DESs composed of diols, glycols, and carbohydrates were found to possess unique solvation interactions, despite the type of HBD being very similar in chemical structure. For example, the [Bet]/7 TEG DES exhibited low hydrogen bond basicity (4.09 ± 0.12) and dispersive-type interactions (0.52 ± 0.01) compared to both the [Bet]/7 (1,4-BD) and [Bet]/7 (1,2-PD) DESs, which was also previously observed for $[Ch^+][Cl^-]$/3 TEG (*a*-term of 4.50 ± 0.12; *l*-term of 0.50 ± 0.01) and $[Ch^+][Cl^-]$/3 (1,4-BD) (*a*-term of 4.91 ± 0.12; *l*-term of 0.53 ± 0.01) DESs.[38] Incorporating the cyclic isosorbide HBD in the [L-Car]/7 Iso DES resulted in significantly lower hydrogen bond basicity (4.14 ± 0.14) and dispersive-type interactions (0.47 ± 0.01) and enhanced dipolarity (2.49 ± 0.08) compared to the [L-Car]/7 (1,4-BD) DES. Given that glycerol contains an additional acidic proton compared to 1,4-BD, 1,2-PD, and TEG, its hydrogen bond acidity (0.71 ± 0.14) was found to be the highest among diols and glycols.

To demonstrate the effect of varying the HBD on the chromatographic retention of analytes, a mixture of 15 probe molecules was separated on the [Bet]/7 (1,4-BD), [Bet]/7 TEG, and [Bet]/7 LvA DESs at 50 °C, as shown in Figure 3. Most probe molecules exhibited the lowest chromatographic retention on the [Bet]/7 (1,4-BD) DES and the highest on the [Bet]/7 LvA DES. While alcohols generally possessed similar retention, naphthalene (analyte 13) and nitrobenzene (analyte 14) retained exceptionally high on the [Bet]/7 TEG DES (15 and 20 min, respectively), compared to 8 and 11 min, respectively, on the [Bet]/7 (1,4-BD) DES. In contrast to diols, alcohol probe molecules retained the lowest on the [Bet]/7 LvA DES due to its weaker hydrogen bond accepting ability. It was also observed that basic probe molecules, such as *N,N*-DMF (analyte 11), retained for almost 31 min on the [Bet]/7 LvA DES compared to under 9 min for both the [Bet]/7 (1,4-BD) and [Bet]/7 TEG DESs, indicating stronger interactions with DESs composed of organic acids. In addition, the analyte elution order was observed to be very different for the 3 DESs, showcasing that some analytes are retained more strongly than others when the HBD is varied.

In order for the solvatochromic dye and IGC methods to be simultaneously used to describe the solvation characteristics of zwitterionic DESs, results from the two approaches should have complete agreement with each other. For the Kamlet–Taft solvent parameters, it was observed that DESs composed of organic acid HBDs offer lower hydrogen bond basicity compared to diols, which is clearly observed with the system constants from the Abraham model. It was also observed that [Bet] DESs composed of the 1,2-PD HBD exhibited slightly higher hydrogen bond accepting ability (β) compared to those composed of 1,4-BD, which generally fit the trend observed for the *a*-term. Additionally, the dipolarity/polarizability (π^*) was found to be statistically similar for all diol and LvA DESs and can also be observed for the *s*-term in the case of [Bet] DESs composed of LvA, 1,2-PD, and 1,4-BD HBDs. Furthermore, the sulfobetaine DESs were generally observed to exhibit lower π^*-term values; an examination of the *s*-term value for the [SB3-BIM]/1.5 CSA DES (1.76 ± 0.06) reiterates that dipolar interactions are the lowest for DESs containing sulfonyl moieties. Clearly, results from the Abraham solvation model

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

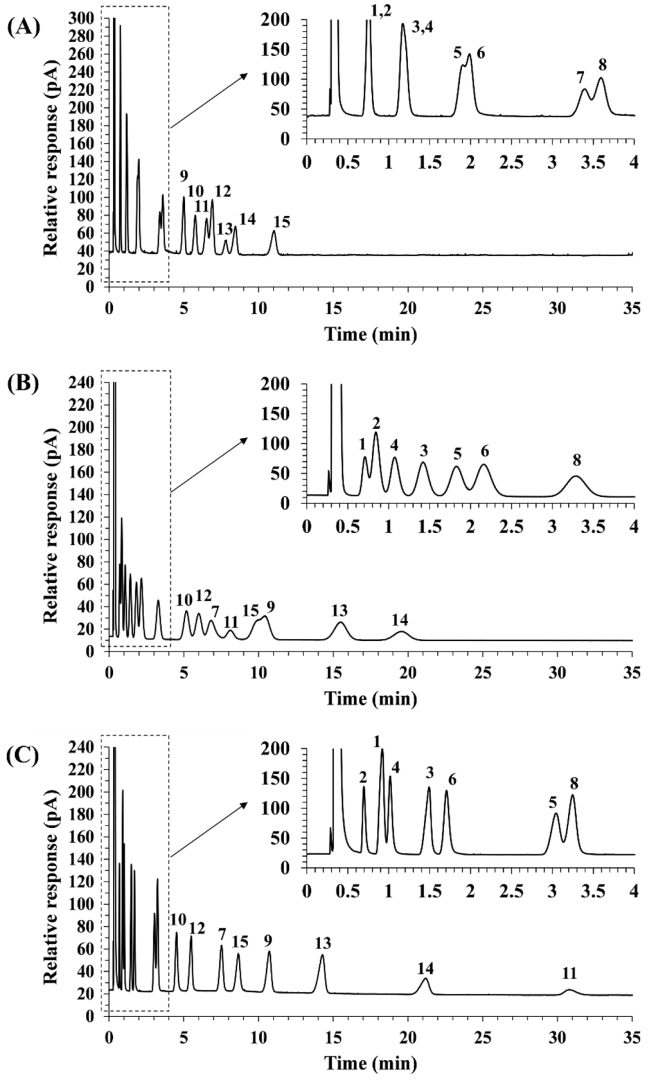


Figure 3. Chromatographic separation of a mixture of 15 probe molecules on the (A) [Bet]/7 (1,4-BD) DES, (B) [Bet]/7 TEG DES, and (C) [Bet]/7 LvA stationary phases at 50 °C. Analytes: 1, ethanol; 2, 1-chlorooctane; 3, 1-bromooctane; 4, 1-propanol; 5, cyclohexanone; 6, 1-butanol; 7, benzaldehyde; 8, 1-pentanol; 9, benzonitrile; 10, cyclopentanol; 11, *N,N*-dimethylformamide (*N,N*-DMF); 12, 1-hexanol; 13, naphthalene; 14, nitrobenzene; and 15, cyclohexanol.

are supported by those of the Kamlet−Taft solvent parameters approach and enable the determination of dispersive-type and n−π/π−π interactions, which could not be achieved using the solvatochromic dyes. Moreover, DESs for which no peak could be detected in the UV−Vis wavelength region for certain dyes could also be studied, as in the case of the [L-Car] DESs composed of Iso and PheAceA HBDs.

3.5. Comparison of Neat HBDs and DESs with Their Respective HBDs. The [Bet] and [L-Car] DESs composed of TEG, 1,4-BD, and 1,2-PD HBDs were compared with neat HBDs to investigate differences in solvation properties between DESs and their neat components. Unlike zwitterionic DESs, neat HBDs do not possess a negatively charged carboxylate or sulfonate functional group that behaves as a strong hydrogen bond base. Consequently, the [Bet]/7 (1,4-BD) (4.99 ± 0.14) and [Bet]/7 (1,2-PD) (4.86 ± 0.12) DESs exhibited higher hydrogen bond basicity compared to 1,4-BD (4.28 ± 0.13) and 1,2-PD (4.56 ± 0.14). Additionally, dipolar interactions for neat HBDs were observed to be lower compared to their DESs. In terms of dispersive-type interactions, the [Bet]/3 (1,4-BD) (0.54 ± 0.01) and [Bet]/3 (1,2-PD) (0.55 ± 0.01) DESs offered much lower interactions compared to 1,4-BD (0.66 ± 0.01) and 1,2-PD (0.67 ± 0.01), while those formed by mixing the HBA and HBD in a 1:7 molar ratio resulted in comparable *l*-term values. Compared to a reported *a*-term value of (4.06 ± 0.11)[42] for TEG at 50 °C, the [Bet]/7 TEG (4.09 ± 0.12) DES possessed very similar hydrogen bond basicity, unlike other diols. Other types of interactions were also found to be similar, with the exception of dispersive-type interactions for the [Bet]/7 TEG (0.52 ± 0.01) DES compared to (0.59 ± 0.01)[42] for TEG. It was also observed that TEG possessed significantly lower hydrogen bond basicity and dispersive-type interactions compared to 1,2-PD and 1,4-BD, while smaller differences in hydrogen bond basicity and dipolar interactions were deduced in the absence of ether linkages.

The neat HBDs and their DESs were further classified into groups based on system constants obtained from the Abraham solvation parameter model. Figure 4 shows the values of

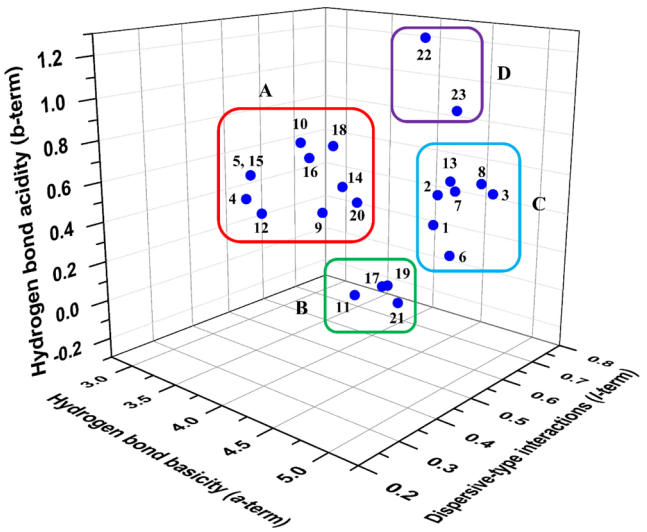


Figure 4. Three-dimensional plot illustrating the relation of hydrogen bond basicity, hydrogen bond acidity, and dispersive-type interactions at 50 °C for over 20 DESs and neat HBDs evaluated in this study and grouped together using k-means clustering. **1**, [Bet]/3 (1,4-BD); **2**, [Bet]/5 (1,4-BD); **3**, [Bet]/7 (1,4-BD); **4**, [Bet]/5 LvA; **5**, [Bet]/7 LvA; **6**, [Bet]/3 (1,2-PD); **7**, [Bet]/5 (1,2-PD); **8**, [Bet]/7 (1,2-PD); **9**, [Bet]/2 PheAceA; **10**, [Bet]/7 PheAceA; **11**, [Bet]/7 TEG; **12**, [Bet]/7 Gly; **13**, [L-Car]/7 (1,4-BD); **14**, [L-Car]/7 Iso; **15**, [L-Car]/7 LvA; **16**, [L-Car]/7 PheAceA; **17**, [4-MM-4-oxide]/PheAceA; **18**, [4-MM-4-oxide]/7 LvA; **19**, [SB3-14]/1.5 CSA; **20**, [SB3-14]/7 LvA; **21**, [SB3-BIM]/1.5 CSA; **22**, 1,4-BD; and **23**, 1,2-PD.

hydrogen bond basicity, hydrogen bond acidity, and dispersive-type interactions being used to construct a three-dimensional plot. Using k-means clustering (see Supporting Information for details), it can be observed that cluster A is entirely composed of DESs made with LvA, PheAceA, and Iso HBDs, while cluster B is predominantly composed of those formed with the CSA HBD. In contrast, cluster C only contains DESs composed of the [Bet] and [L-Car] HBAs and 1,4-BD and 1,2-PD HBDs, while cluster D is exclusively composed of neat HBDs. It can be seen from the plot that DESs composed of

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Ind. Eng. Chem. Res. 2023, 62, 4084−4094

organic acids and diols tend to form their own separate clusters while CSA and LvA, that differ in their pKa strengths (1.2 and 4.64, respectively), do not fall under the same cluster thus, further exemplifying that the HBD has a stronger influence on solvation interactions of zwitterionic DESs compared to the HBA. In addition, Figure S3 provides a normalized ternary plot comprising hydrogen bond basicity, dipolar interactions, and dispersive-type interactions for 23 DESs and neat HBDs to further demonstrate the variations in solvation properties of zwitterionic DESs.

4. CONCLUSIONS

DESs composed of zwitterionic HBAs form a popular class of sustainable solvents. Given that zwitterionic salts can form DESs with a wide variety of HBDs, these solvents possess unique and diverse physicochemical properties due to their amphiphilic nature. While zwitterionic DESs have been employed as solvents or reaction media in catalysis, organic synthesis, and chemical separations, their solvation characteristics have never been thoroughly studied with either the solvatochromic dye method or IGC.

Using the Nile red dye polarity scale, glycerol-based DESs were observed to possess the highest polarity among the DESs examined in this study, while those composed of organic acids were generally more polar than most diols. Studying the same DESs using Kamlet−Taft solvent parameters revealed that DESs composed of glycerol possess stronger hydrogen bond donating ability than most DESs while the α-term varied significantly for those composed of organic acids when the molar ratio of HBD was varied, unlike diols. Additionally, the [SB3-14] HBA was found to offer the highest hydrogen bond donating ability compared to [Bet], [L-Car], and [4-MM-4-oxide] HBAs. Those DESs that exhibited higher hydrogen bond donating ability generally offered lower hydrogen bond accepting and vice versa, while dipolar interactions did not vary drastically when systematic modifications were made to the DES chemical structure. Using the Abraham solvation parameter model, up to five types of unique interactions could be compared for the same DESs, and those composed of sulfobetaine HBAs possessing longer alkyl chains generally exhibited higher hydrogen bond basicity and dispersive-type interactions. Solvation models generated using both the solvatochromic dye approach and IGC were generally in agreement, and it was found that the type of HBD more strongly influenced the solvation interactions of zwitterionic DESs compared to the nature of the HBA. In addition, the solvation interactions of neat HBDs were generally observed to be similar to their DESs only at very high HBA/HBD ratios of 1:7. Results from this study highlight the need to use multiple approaches and polarity scales to elucidate the solvation properties of complex solvents and provide the framework to simultaneously employ IGC and fluorescent dye methods for their complete characterization.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.iecr.2c04425.

List of chemicals and reagents, UV−vis absorption spectra of solvatochromic dyes, description of k-means clustering, equations employed to calculate the E_{NR} values, Kamlet−Taft solvent parameters, system constants using the Abraham solvation parameter model, water content of HBDs and their DESs, viscosity measurements obtained on DESs and neat HBDs, melting point/glass-transition temperatures of DESs, list of probe molecules and corresponding solute descriptors, and ^{1}H and ^{13}C NMR spectra for DESs evaluated in the study (PDF)

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Notes

The authors declare no competing financial interest.

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